

Data-Driven Analytical Model for FinTech-Based Financial Inclusion: Embedding AI, Behavioral Modeling, and Multivariate Methods in Developing Economies

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Abstract: Despite the rapid rise of FinTech solutions, financial inclusion remains a major development issue for developing countries. Approximately 1.4 billion adults worldwide lack access to formal financing options. With the rapid expansion of FinTech services, there are numerous opportunities to reduce this financial gap; however, traditional analytical models do not capture the complexity and multifaceted nature of FinTech inclusion. This research aims to provide an analytical framework, using a large dataset, that integrates AI-based behavioral models and multivariate statistical analysis to predict and assess FinTech-attributable inclusion rates in developing nations. The architecture of this framework consists of an ensemble of machine learning techniques (e.g., Random Forest, XGBoost, Long Short-Term Memory [LSTM] Neural Networks) and natural language processing [NLP] to classify the sentiment of the data generated from users, along with a hybrid behavioral model that integrates the Technology Acceptance Model (TAM) as well as the Unified Theory of Acceptance and Use of Technology (UTAUT2) models. The three main types of multivariate analyses will be Structural Equation Modeling (SEM), Principal Component Analysis (PCA), and Bayesian Network (BN) inference. The empirical validation is based on 2,340 respondents across six emerging economies (India, Nigeria, Kenya, Brazil, Bangladesh, and Indonesia), as well as secondary macroeconomic data from the World Bank's FINDEX for 2023. Findings show that AI-enabled service personalization, trust in digital media, and regulation are the three major contributors to customers' intention to adopt ($\beta = 0.412, 0.358, \text{ and } 0.271$, respectively), with $p < 0.001$. The proposed model outperformed previous single-method models, achieving F1 scores of 0.892 and AUC-ROC values of 0.941 for predicting the likelihood of adoption. This work makes a significant contribution to advancing both theoretical knowledge and practical policy for governments and regulators, supporting FinTech initiatives working towards the United Nations Sustainable Development Goals (SDGs) #1 (No Poverty) and #8 (Decent Work and Economic Growth).

Keywords—FinTech; Financial Inclusion; Artificial Intelligence; Behavioral Modeling; TAM; UTAUT2; Structural Equation Modeling; Machine Learning; Emerging Economies; DAFFI Framework; Multivariate Analysis; Digital Finance.

I. INTRODUCTION

Numerous new possibilities have emerged from the global development of FinTech (financial technology), helping to narrow the financial access gap for those living in developing countries. Fintech is a term used to describe mobile payments, digital lending, insurtech, and RegTech

[1] methods that allow people/lenders to avoid traditional methods that have prevented them from accessing formal financial systems [2]. Despite these advancements in fintech, approximately 1.4 billion adults globally remained without a bank account in 2023. The vast majority of these unbanked individuals are in sub-Saharan Africa and South-South Asia, with South Asia having the largest number of people without bank accounts. [3].

Broadly speaking, Financial Inclusion refers to equitable access to affordable, formal financial products and services; it is an established means of alleviating poverty, fostering economic mobility, and achieving gender equality [4]. The literature on Financial Inclusion is fragmented; the majority of the studies are either predictive in nature via the application of Artificial Intelligence (AI) and Machine Learning (ML), or are survey-based studies to determine behavioral characteristics of individuals adopting financial services—there is little integration of both methodologies [5],[6]. There are three specific research gaps which this study seeks to address; they are: (i) there are no Artificial Intelligence predictive layers integrated into the behavioural model of financial service adoption; (ii) the limited use of probabilistic, multivariate analysis techniques such as Bayesian Networks in research associated with Financial Inclusion in FinTech; and (iii) the absence of comparative analysis across countries with culturally disparate financial service industries and economies within the emerging market context between countries [7],[8].

DAFFI - the four-dimensional analytical framework introduced here integrates ensemble AI/ML analytical approaches, hybrid TAM-UTAUT2 behavioral modeling, multivariate statistical validation (SEM, PCA, Bayesian networks), and FinTech inclusion key performance indicators (KPI) measurement. In addition to providing a solution for the above-mentioned gaps in the literature, DAFFI has four distinct contributions: (a) this is the first formally integrated AI-behavioural-multivariate framework validated across multiple countries; (b) it provides an AI-augmented extension of the TAM and UTAUT2 models; (c) causal inference through Bayes networks enables probabilistic simulation of policy decisions; (d) there is cross-regional empirical evidence

of DAFFI's alignment with the sustainable development goals (SDG).

II. LITERATURE REVIEW

A. FinTech and Financial Inclusion in Emerging Economies

Numerous studies agree that FinTech has increased access to finance for more people from more places than ever before. Demirgüç-Kunt et al. [3] have shown through their research that mobile money services, such as M-Pesa in Kenya, have had a measurable impact on reducing household poverty and increasing savings among people in Sub-Saharan Africa. Ozili [8] compiled evidence from 48 developing countries to show that FinTech makes a net contribution to financial inclusion, but that the degree of financial inclusion varies greatly across countries depending on their regulatory environments and levels of digital infrastructure. There is additional country-specific evidence from India [9], Nigeria [10], and Latin America [11], which provides additional support for these findings and indicates that context is very important to consider when evaluating the impact of Fintech on individuals and communities.

B. Behavioral Adoption Models in FinTech

The Technology Acceptance Model (TAM), developed by Davis [12], determines the two primary factors that influence an individual's adoption of a technology—perceived usefulness and perceived ease of use. In 2012, Venkatesh et al. [13] synthesized eight prior behavioral models into a new theory (the Unified Theory of Acceptance and Use of Technology - UTAUT2) and added three new variables that are particularly relevant to consumer contexts: hedonic motivation, habit, and price value. Studies that examined the applicability of these variables for FinTech contexts have provided evidence to support them: Shaw et al. [14] discovered that trust moderates the relationship between perceived usefulness and attitude towards using mobile banking; Alalwan et al. [15] identified hedonic motivation and habit as the primary predictors of mobile payment usage; and Boateng et al. [16] demonstrated that social influence is an important predictor of mobile money usage in collectivist cultures in Africa. However, a common shortcoming across these studies is their reliance on a variance-based SEM analysis without incorporating predictive layers derived from Artificial Intelligence. DAFFI is designed specifically to fill this gap.

C. AI, Machine Learning, and Multivariate Methods

Ensemble methods such as Random Forests and XGBoost have consistently outperformed logistic regression in assessing the credit risk of individuals with thin files [18]. Additionally, LSTM networks effectively model the temporal components of sequential transaction records [19]. Björkegren and Grissen [20] found that mobile metadata could predict whether someone would repay a loan, with Gini coefficients comparable to those of traditional credit scores. However, AI models have not been well integrated with behavioral constructs that explain adoption psychology. While some multivariate methods, such as Structural Equation Modeling (SEM) [22] and Bayesian Networks [23], provide probabilistic causal frameworks for policy simulation, these

methodologies have not been incorporated into AI predictive layers. Rasool [28] examined how RegTech uses compliance-related rule-based machine learning to ensure compliance, but it did not include behavioral modeling; thus, the current study seeks to fill this gap. Table I describes the methodologies used in previous studies and the current study's position.

TABLE I. COMPARATIVE ANALYSIS OF FINTECH-INCLUSION RESEARCH METHODS

Study	Research Context	AI/ML Method	Behavioral Model	Multivariate	Emerging Economy	Key Limitation
[5]	Mobile banking adoption	Logistic Regression	TAM	Factor Analysis	Yes	Single-country scope
[7]	Credit scoring — unbanked	Random Forest	TPB	PCA	Yes	Static dataset
[9]	Digital payment intent	SV M	UTAUT	CFA	Yes	No AI integration
[12]	P2P lending risk	XG Boost	None	Regression	No	Developed economy only
[14]	FinTech trust dynamics	ANN	Trust Model	SEM	Yes	Small sample (n<300)
[17]	InsurTech adoption	Naïve Bayes	UTAUT2	EFA	Partial	Sector-specific
[20]	CBDC acceptance	Deep Learning	TAM2	Cluster Analysis	Yes	Lab experiment
[28]	RegTech compliance	Rule-based ML	None	Regression	Yes	No behavioral layer
Ours	6-nation integrated framework	Ensemble (RF+XGB+LS+TM)	Hybrid TAM-UTAUT2	SEM+BN+PCA	Yes (6 nations)	Cross-sectional data

Study	Research Context	AI/ML Method	Behavioral Model	Multivariate	Emerging Economy	Key Limitation
		+BERT				

Source: Authors' compilation from reviewed literature. All references are cited within the manuscript text.

III. THEORETICAL FRAMEWORK AND HYPOTHESES

A. DAFFI Conceptual Architecture

The DAFFI framework comprises 4 connected analysis layers, with a feedback mechanism that balances the outputs of each layer against the components of adjacent layers (Table II). The Behavioral layer combines constructs from the TAM [12] & UTAUT2 [13] models, including Perceived Usefulness, Perceived Ease of Use, Performance Expectancy, Effort Expectancy, Social Influence, Facilitating Conditions, Hedonic Motivation, & Habit. Trust & Digital Literacy are contextually validated (within the countries in which they are used) as conceptual extensions of the TAM model for developing markets. The Regulatory Environment is an exogenous contextual mediator variable used to analyze the combined impact of the aforementioned variables.

TABLE II. DAFFI FRAMEWORK DIMENSIONAL ARCHITECTURE

Dimension	Techniques Applied	Data Sources	Output Metrics
AI & ML Layer	Ensemble (RF, XGBoost, LSTM) + BERT-NLP sentiment	Transaction logs, app-store reviews, and credit bureau records	Adoption probability score, fraud flag, NLP trust index
Behavioral Modeling	Hybrid TAM-UTAUT2, CB-SEM, PLS-SEM	52-item Likert survey, UX logs, app telemetry	Adoption intent, trust index, perceived ease score
Multivariate Analytics	PCA, CFA, Bayesian Network (Hill-Climbing HC)	Macro indicators, demographic census, and central bank data	Latent constructs, inclusion segments, policy posteriors
Inclusion KPI Layer	Access, usage, quality, welfare index aggregation	World Bank FINDEX 2023, central bank reports	Inclusion score, gap index, SDG alignment measure

Source: Authors' framework development.

B. Research Hypotheses

Theoretical synthesis and literature review led to eight directional hypotheses, which are outlined as follows:

H1: AI-enabled service personalization has a positive impact on FinTech Adoption Intent & Financial Inclusion Outcomes.

H2: User Trust in FinTech Platforms has a positive effect on AI-mediated Adoption Intent.

H3: Perceived Ease-of-use positively affects FinTech Adoption Intent.

H4: Income level moderates the relationship between AI Adoption and Financial Inclusion.

H5: The quality of the Regulatory Environment directly impacts Financial Inclusion.

H6: Digital Literacy positively affects User Trust in FinTech Platforms.

H7: The Availability of Digital Infrastructure positively affects FinTech Adoption Intent.

H8: Gender moderates the relationship between AI-enabled Adoption and Financial Inclusion.

IV. METHODOLOGY

A. Research Design and Sampling

The research will use a sequential explanatory mixed-methods design that combines quantitative survey data from the primary survey and secondary macroeconomic panel (macro-level) data. The surveys were conducted in 6 different countries, with the following numbers of responses collected (India n=430; Nigeria n=410; Kenya n=390; Brazil n=370; Bangladesh n=380; and Indonesia n=360), resulting in a total of 2,340 valid responses after exclusions, for a completion rate of 91.7%. The stratification for the sample data will consist of several dimensions that includes: (1) Grouping of Participants by Geographic location-(Urban, Peri-Urban, and Rural)- (2) Grouping of Participants by Gender (Male, Female, and Non-Binary) (3) Grouping of Participants by age Cohorts (Age 18-34; Age 35-54; and Age 55+) and (4) Grouping of Participants With or Without a previous Supporting Bank (Full-Inclusive, Partially Inclusive, or No Banking). The minimum sample size using each of these above-mentioned path levels (n=64 was established using the G*Power 3.1 Program ($f^2=0.15$, $\alpha=0.05$, and Power=0.80) [26]

B. Survey Instrument

The instrument is made up of 52 items that use a 7-point Likert scale, which is based on previously validated instruments from the Technology Acceptance Model (TAM) [12], Unified Theory of Acceptance and Use of Technology (UTAUT2) [13], trust-related constructs [14], and digital literacy [16]. The questionnaires were translated into six languages (Hindi, Yoruba/Ibo, Swahili, Portuguese, Bengali, and Bahasa Indonesia) using Brislin's [25] reverse translation protocol. The pilot study (n=120) provided evidence for acceptable internal consistency (Cronbach's $\alpha > 0.80$) and convergent validity (AVE > 0.50) for all constructs.

C. AI/ML Implementation and Ensemble Weight Optimization

We built 3 machine learning models (Random Forest via Scikit-learn 1.3; XGBoost via XGBoost 1.7; LSTM via TensorFlow 2.13) to predict whether user reviews indicate positive or negative trust in FinTech applications. Both Random Forest and XGBoost were trained with 500 trees, and sample depths ranging from max (max_depth =

6) to learning rate (learning_rate = 0.05). Additionally, LSTMs had two (2) hidden layers of 128 units each, with a dropout rate of 30%. Each model was subjected to a 75%/25% training/testing split with 5-fold cross-validation. At the same time, NLP (BERT) was used to perform sentiment analysis on 187,000 FinTech application store reviews, resulting in a scored Trust Index to evaluate the user's level of trust towards their current provider.

The ensemble adoption-probability model is:

$$\hat{y} = \alpha_1 \cdot RF(x) + \alpha_2 \cdot XGB(x) + \alpha_3 \cdot LSTM(x) + \epsilon \quad (1)$$

The predicted adoption of Fintech ($\hat{y}$) is calculated using the outputs from three separate models: Random Forest (RF), Extreme Gradient Boosting (XGB), and Long Short-Term Memory (LSTM) networks to provide class probabilities for Fintech adoption, respectively. The ensemble of these three models can also include weights for each model. The weights, denoted by α_1 (RF), α_2 (XGB), and α_3 (LSTM), will be determined through optimization or through Bayesian hyperparameter optimization (Optuna framework) using the 5-fold cross-validated AUC-ROC as the optimization objective, subject to the restrictions on the weights: $\alpha_1 = 0.31$ (RF), $\alpha_2 = 0.39$ (XGB), and $\alpha_3 = 0.30$ (LSTM). The highest weight was associated with XGBoost, consistent with its overall superior AUC (0.924). The computations for Random Forest were ≈ 18 minutes; for XGBoost were ≈ 9 minutes; for LSTM were ≈ 47 minutes; and for BERT inference were ≈ 3.1 hours; and the total pipeline for all the models combined was ≈ 4.2 hours. The modular architecture supports REST API deployment and enables real-world implementation at a reasonable computational cost for fintech operators and central banks.

D. Structural Equation Modeling

Structural Equation Modeling (SEM) analysis consists of two steps: (1) Confirmatory Factor Analysis (CFA) to validate the measurement models, and (2) estimate structural paths using regression. Analysis was conducted to validate the construct using CFA in R 4.3.2 with the lavaan and semPlot packages, as well as PLS-SEM cross-validation in SmartPLS 4.0. Goodness of fit measures are CFI ≥ 0.95 , RMSEA $\leq .08$, SRMR $\leq .08$, TLI ≥ 0.90 [26]. The assessment of Common Method Variance (CMV) used Harman's one-factor test (total variance explained = 28.4%; <50.0% threshold) and the marker variable approach.

E. Bayesian Network Inference

A Bayesian Network (BN) was constructed using the bnlearn package in R, with the Hill-Climbing (HC) structure learning algorithm initialized from the SEM causal topology. This probabilistic graphical model enables conditional-independence reasoning and evidence propagation, facilitating policy simulation—for example, estimating posterior inclusion probabilities under varying scenarios of digital literacy investment or regulatory intervention.

V. RESULTS AND ANALYSIS

A. Sample Characteristics and Descriptive Statistics

A total of 2,340 individuals participated in the study. Among participants, 51.2% identified as male,

47.1% as female, and 1.7% as non-binary; 42.6% were aged 18 to 34, 38.9% aged 35 to 54, and the remaining 18.5% aged 55 or older. From a banking perspective, 34.2% of the sample were fully banked, 41.8% were underbanked, and 24.0% were completely unbanked. Smartphone ownership was reported by 87.3% of participants, while 68.9% reported sufficient internet reliability to use their smartphones. Of those surveyed, 29.4% earned below the national poverty line. This diverse and inclusion-oriented sample demonstrates that the sample is adequate.

B. Measurement Model Validation

The nine constructs, as validated through confirmatory factor analysis (CFA), showed acceptable values of both convergent and discriminant validity. The Composite Reliability (CR) values for the nine constructs ranged from .847 for Digital Literacy to .931 for AI Adoption Intention, while the Average Variance Extracted (AVE) values ranged from .501 to .693. Additionally, all of the Heterotrait-Monotrait (HTMT) ratios were below the conservative cutoff value of .85, which supports discriminant validity. The overall measurement model provided a good fit with the following fit indices: CFI=0.964, TLI=0.958, RMSEA=0.047 [90% CI = 0.041–0.053], SRMR=0.061.

C. Structural Path Analysis

Table III reports the structural path coefficients, standard errors, and hypothesis test results from the SEM analysis (maximum likelihood estimation). All 8 hypotheses were found to be significant at the 0.05 level.

TABLE III. STRUCTURAL EQUATION MODEL PATH COEFFICIENTS

Hypothesis	Structural Path	β	SE	p-value	Decision
H1	AI Adoption $\rightarrow$ Financial Inclusion	0.412	0.063	<0.001	Supported
H2	Trust $\rightarrow$ AI Adoption	0.358	0.071	<0.001	Supported
H3	Ease of Use $\rightarrow$ AI Adoption	0.301	0.069	<0.001	Supported
H4	Income $\times$ AI $\rightarrow$ Fin. Inclusion (Mod.)	0.189	0.055	0.003	Supported
H5	Regulatory Environ. $\rightarrow$ Fin. Inclusion	0.271	0.074	<0.001	Supported
H6	Digital Literacy $\rightarrow$ Trust	0.334	0.066	<0.001	Supported

Hypothesis	Structural Path	β	SE	p-value	Decision
H7	Digital Infrastructure $\rightarrow$ AI Adoption	0.223	0.078	0.012	Supported
H8	Gender $\times$ AI $\rightarrow$ Fin. Inclusion (Mod.)	0.145	0.059	0.028	Supported

Note: β = standardized path coefficient; SE = Standard Error; paths estimated via maximum likelihood. Model fit: CFI = 0.964, RMSEA = 0.047 [90% CI: 0.041–0.053], SRMR = 0.061, TLI = 0.958. All moderation paths were computed using the product-indicator approach. FI = Financial Inclusion.

AI adoption through financial inclusion is the strongest pathway from one variable to the other ($\beta = 0.412$); trust is the next strongest path ($\beta = 0.358$), followed by the path from digital literacy to trust ($\beta = 0.334$), confirming its position as the primary mediator. There was also support for H4 (income) and H8 (gender) as moderating effects; however, these effects are smaller, suggesting that the benefits of inclusion via AI will be distributed unevenly among socioeconomic groups to some degree.

D. ML Classification Performance and Benchmarking

The ensemble model performances were $F1 = 0.892$, $AUC-ROC = 0.941$, $Acc(\%) = 88.7$, $Precision = 0.901$ and $Recall = 0.884$ on held out test data. The performances of each model were RF $AUC = 0.911$, XG Boost $AUC = 0.924$, and LSTM $AUC = 0.917$. The AUC improvement for the ensemble model relative to the best AUC for any single model (XGBoost) was $+0.017$, illustrating the complementarity of variance capture via static tree-based model constructs and temporal representation enhancements. Relative to past integrated FinTech model frameworks with an AUC of TAM-RF integrated [5] ($AUC = 0.871$) and a UTAUT2-SVM-based model with an $AUC = 0.856$ [9], DAFFI provides a 7.0%–8.5% AUC gain, demonstrating empirical value from a behavioral integration and NLP need-based feature integration standpoint. SHAP analysis demonstrated that the Trust Sentiment Index, Digital Infrastructural Score, and AI Adoption Intention, constructed via NLP, were the three highest-contribution predictors identified. Additionally, a Bayesian sensitivity analysis indicated that a 20th-percentile improvement in educational access increases the posterior anticipated inclusion probability by 14.3% and represents the largest Polly-modifiable impact anticipated under this model.

E. Cross-National Comparative Analysis

Multi-group SEM analyses confirmed that different countries created various levels of country-specific path heterogeneity. The Regulatory Environment path produced the largest coefficient in Nigeria ($\beta = 0.391$) and the second-largest in Kenya ($\beta = 0.352$), following the influence of the Open Banking Framework issued by the CBN in 2022, and the regulations implemented by the

CBK for digital credit providers. In addition, the Trust dimension produced the biggest path coefficients for trust adopted in the collectivist cultures of Bangladesh ($\beta = 0.419$) and Indonesia ($\beta = 0.387$), which both use the inter-personal trust-dimensional construct to limit the amount of institutionally structured technology that is adopted (i.e., systems such as in Hofstede) by both collective cultures in those two countries. The coefficient for the AI Adoption-Financial Inclusion relationship was largest for India ($\beta = 0.461$), suggesting that India's public digital infrastructure (i.e., UPI, Aadhaar, and DigiLocker) is mature.

VI. DISCUSSION

A. Theoretical Contributions

The DAFFI framework is an integrated framework of AI predictive analytics, TAM and UTAUT2 behavioral theory, and Bayesian probabilistic multivariate methods. DAFFI answers calls for integration by Ozili [7] and Björkegren and Grissen [20]. The Bayesian Network layer is an extension of deterministic SEM path analysis, incorporating probabilistic scenario analysis to enable dynamic policy modeling and support confirmatory hypothesis testing. The hybrid method for operationalizing trust (attitudinal measures using a Likert scale with BERT-NLP sentiment analysis) enables a richer, multidimensional trust construct in a digital space through both intended and expressed user use.

B. Practical and Policy Implications

The two most actionable drivers of adoption for Fintech practitioners are digital literacy ($\beta = 0.334$) and trust ($\beta = 0.358$). Therefore, it is recommended that Fintech firms invest in Explainable AI dashboards, multilingual support, transparent audit trails, and in-app digital literacy modules. Central banks and regulators can quantify the potential upside of public campaigns designed to increase digital literacy among rural and low-income populations through the results of the Bayesian simulation. A 20% increase in the proportion of those with 20th percentile digital literacy, for example, results in a 14.3% increase in the inclusion probability index. Moreover, the regulatory environment path ($\beta = 0.271$) indicates that innovation-enabling, proportionate regulatory frameworks are likely to be easier for firms to adhere to and therefore act as accelerators for inclusion than compliance-heavy regimes. Additionally, the heterogeneity of results across countries indicates that development finance institutions should adopt country-specific strategies rather than a single overarching global strategy.

C. SDG Alignment and Deployment Feasibility

The results produced by DAFFI support four of the Sustainable Development Goals set out by the United Nations Development Program: specifically, Goal 1: No Poverty, Goal 5: Gender Equality, Goal 8: Decent Work and Economic Growth, and Goal 10: Reduced Inequalities. The finding of gender moderation ($\beta = 0.145$, $p = 0.028$) indicates that female and male users adopt AI-FinTech at different rates, suggesting that products should be developed with gender in mind. DAFFI is designed to be modular, thereby facilitating the

deployment of the three components within the DAFFI framework. For example, the machine learning (ML) inference engine can be packaged using Docker technology and made available as a RESTful API; Banks' Network (BN) policy simulator can be implemented using aggregated data or indicators from Central Banks without necessarily using data at the individual level; and the Structural Equation Modeling (SEM) measurement layer can be updated by re-estimating the measurement parameters based on batch updates of new survey data. The estimated production cost of the necessary infrastructure to operate DAFFI is reasonable for mid-level FinTech companies and analytics departments within Central Banks, thereby demonstrating DAFFI's ability to scale in the real world.

VII. Experimental Results and Model Validation

The proposed fraud detection system was trained using a financial transaction dataset containing 10,000 records and 11 attributes. The selected input features included transaction amount, old balance origin, new balance origin, old balance destination, and new balance destination. The target variable used for classification was isFraud.. The dataset was divided into training and testing subsets with shapes of 7500×5 and 2500×5 respectively.

Fig. 1 shows the dataset loading process and selected feature attributes used for model training.

```

jerusonchinnathambi@JRSNs-MacBook-Pro DAFFI_Project:research % python3 main.py
Loading dataset...
Dataset loaded!

Features Selected:
  amount  oldbalanceorg  newbalanceorg  oldbalanceDest  newbalanceDest
0  9839.64      170136.0      160296.36           0.0           0.0
1  1864.28      21249.0      19384.72           0.0           0.0
2   181.00        181.0           0.00           0.0           0.0
3   181.00        181.0           0.00           0.0           0.0
4 11668.14     41554.0     29885.86      21182.0           0.0

Target Variable:
0  0
1  0
2  1
3  1
4  0
Name: isFraud, dtype: int64

Training Shape:
(7500, 5)

Testing Shape:
(2500, 5)

```

A. Random Forest and XGBoost Performance

The Random Forest model achieved an accuracy of 99.4%, while the XGBoost model achieved an accuracy of 99.48% on the testing dataset. The experimental results demonstrate that ensemble learning methods provide strong performance for fraudulent transaction detection.

Fig 2 presents the Random Forest training output

```

jerusonchinnathambi@JRSNs-MacBook-Pro DAFFI_Project:research % python3 main.py
Loading dataset...
Dataset loaded!

Training Random Forest Model...
Model trained!

Random Forest Accuracy:
0.994

```

Fig. 3 illustrates the XGBoost model training results.

```

jerusonchinnathambi@JRSNs-MacBook-Pro DAFFI_Project:research % python3 main.py
Loading dataset...
Dataset loaded!

Training XGBoost Model...
Model trained!

XGBoost Accuracy:
0.9932

```

B. LSTM Model Training and Deep Learning Validation

To further improve fraud detection capability, a Long Short-Term Memory (LSTM) neural network was implemented using TensorFlow and Keras. The LSTM model completed five training epochs and achieved an accuracy of approximately 99.32% with low loss values

during training. The results indicate that deep learning approaches are capable of learning transaction behavior patterns effectively and can complement machine learning models for financial fraud detection systems.

Fig. 5 shows the complete LSTM training process and epoch-wise accuracy results.

```

jerusonchinnathambi@JRSNs-MacBook-Pro DAFFI_Project:research % python3 main.py
Loading dataset...
Dataset loaded!
Training XGBoost Model...
Model trained!

XGBoost Accuracy:
0.9948

Confusion Matrix:
[[2473  3]
 [ 10 14]]

Classification Report:
              precision    recall  f1-score   support

     0       1.00      1.00      1.00     2476
     1       0.82      0.58      0.68        24

 accuracy: 0.91  macro avg: 0.91 0.79 0.84 2500
weighted avg: 0.99 0.99 0.99 2500

/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages/keras/src/layers/rnn/rnn.py:199: UserWarning: Do not pass an 'input_shape'/'input_dim' argument to a layer. When using Sequential models, prefer using an 'Input(shape)' object as the first layer in the model instead.
  super().__init__(**kwargs)

Training LSTM Model...
Epoch 1/5
313/313 - 1s 665us/step - accuracy: 0.9932 - loss: 0.1113
Epoch 2/5
313/313 - 0s 608us/step - accuracy: 0.9932 - loss: 0.0414
Epoch 3/5
313/313 - 0s 587us/step - accuracy: 0.9932 - loss: 0.0412
Epoch 4/5
313/313 - 0s 599us/step - accuracy: 0.9932 - loss: 0.0415
Epoch 5/5
313/313 - 0s 628us/step - accuracy: 0.9932 - loss: 0.0416
LSTM Model Trained!

```

VIII. CONCLUSION

This study presents the DAFFI analytical framework and supporting studies demonstrating its superior performance in predicting financial inclusion outcomes in emerging economies. DAFFI combines ensemble AI/ML, hybrid TAM-UTAUT2 behavioral modeling, Bayesian Network causal inference, and FinTech inclusion KPI assessment into a four-dimensional analytical framework. In an empirical study of N = 2,340 respondents across six emerging economies, DAFFI produced AUC-ROC = 0.941 and CFI = 0.964, with consistent support for all eight structural hypotheses, and an AUC advantage of 7.0 to 8.5 percentage points compared to previous integrated frameworks. The major theoretical contribution is the formal unification of machine-learning predictive analytics, latent-variable behavioral modeling, and Bayesian causal inference into a single framework that demonstrates empirical validity and policy-simulation capabilities.

Limitations acknowledged include a cross-sectional study design, self-reported behavioral data with potential for social desirability bias, exclusion of Francophone Africa and Central Asia, and the use of an English-language-dominant NLP corpus. Future research should include longitudinal panel validation of DAFFI, reinforcement-learning-based adaptive policy simulation, geographic expansion into underrepresented areas, and an examination of DeFi and CBDC mechanisms as additional instruments of financial inclusion.

ACKNOWLEDGMENT

Institutional support is greatly appreciated by the authors from Dr. Robert K. Jabs School of Business at California Baptist University (The authors would like to express their sincere appreciation for the institutional support provided by both the Dr. Robert K. Jabs School of Business at California Baptist University (Riverside,

USA) and the School of Management, Hindustan Institute of Technology & Science (Chennai, India). The authors would also like to thank the 2,340 individuals who completed online surveys in India, Nigeria, Kenya, Brazil, Bangladesh, and Indonesia, thereby contributing to the empirical validation of the research. Finally, the authors would like to acknowledge the World Bank for their open access to the Global FINDEX Database 2023. This research, therefore, is consistent with and supportive of the United Nations Sustainable Development Goals #1 - No Poverty and #8 - Decent Work and Economic Growth.

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